

Yating Wu

Ph.D. Candidate

LLM Agents, LLM Reasoning, AI for Science

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Summary

I develop LLM reasoning and agent methods for structuring evidence, tracking information flow, and supporting scientific discovery across language, multimodal data, and tools.

- 🏆 **Selected recognitions:** MIT EECS Rising Stars; EMNLP 2024 Outstanding Paper Award (Top 0.4%).
- 💡 **Pretraining/Post-training:** astronomy-related foundation models, learning signals, and evaluation methods for LLM/VLM reasoning.
- 💡 **LLM Agents:** long-context memory, planning, tool use, and dependency-aware context.
- 💡 **LLM Reasoning:** reasoning structure, information flow, and evidence grounding across text, figures, and multimodal outputs via Questions Under Discussion (QUD).

Education

- 2022 – Present **Ph.D. Candidate in Electrical and Computer Engineering**, *The University of Texas at Austin*
Advisors: Junyi Jessy Li, Alexandros G. Dimakis
- 2020 – 2022 **M.S. in Electrical and Computer Engineering**, *The University of Texas at Austin*
- 2014 – 2019 **B.Eng. in Computer Science & B.A. in Japanese**, *Dalian University of Technology*
- 2017 - 2018 **Exchange student in Computer Science**, *The University of Tokyo*
Advisor: Toshihiko Yamasaki

Work Experience

- Jun - Oct. 2025 **Applied Scientist Intern**, *AWS AI Labs*, Santa Clara, CA
 - Built long-context memory support for LLM coding agents, including dependency-structured context construction for multi-step tool use
- May - Aug. 2024 **Applied Scientist Intern**, *Amazon Alexa AGI*, Sunnyvale, CA
 - Enhanced planning and tool-use capabilities in LLM agents through question-guided reasoning and agent evaluation

Papers Under Submission and Preprints

- [1] **Yating Wu**, Yuhao Zhang, Sayan Ghosh, Sourya Basu, Anoop Deoras, Jun Huan, Gaurav Gupta “ContextWeaver: Selective and Dependency-Structured Memory Construction for LLM Agents.” Under Submission, 2026. [\[Paper\]](#)

- [2] **Yating Wu**, William Rudman, Venkata S Govindarajan, Alexandros G. Dimakis, Junyi Jessie Li “Multimodal QUD: Inquisitive Questions from Scientific Figures.” Under Submission, 2026. [\[Paper\]](#)

Publications

- [3] Ramya Namuduri, **Yating Wu**, Anshun Asher Zheng, Manya Wadhwa, Greg Durrett and Junyi Jessie Li “QUDsim: Quantifying Discourse Similarities in LLM-Generated Text.” Conference on Language Modeling (**COLM**), 2025 [\[Paper\]](#)
- [4] **Yating Wu***, Ritika Rajesh Mangla*, Alexandros G. Dimakis, Greg Durrett, Junyi Jessie Li. “Which questions should I answer? Salience Prediction of Inquisitive Questions.” Conference on Empirical Methods in Natural Language Processing (**EMNLP Oral**), 2024. **Outstanding Paper Award (Top 0.4%)**. [\[Paper\]](#)
- [5] Negin Raoof*, **Yating Wu***, Carlos Bonilla*, Junyi Jessie Li, Stephanie M Grasso, Alex Dimakis, Zoi Gkalitsiou. “Modeling Bilingual Disfluencies with Large Language Models.” Workshop on LLMs and Cognition in International Conference on Machine Learning (**ICML Workshop**), 2024. [\[Paper\]](#)
- [6] **Yating Wu**, Ritika Rajesh Mangla, Greg Durrett, Junyi Jessie Li. “QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing.” Conference on Empirical Methods in Natural Language Processing (**EMNLP Oral**), 2023. [\[Paper\]](#)
- [7] **Yating Wu***, William Sheffield*, Kyle Mahowald, and Junyi Jessie Li. “Elaborative Simplification as Implicit Questions Under Discussion.” Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2023. [\[Paper\]](#)
- [8] Wei-Jen Ko, **Yating Wu**, Cutter Dalton, Dananjay Srinivas, Greg Durrett and Junyi Jessie Li. “Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion.” Findings of the Association for Computational Linguistics (**ACL**), 2023. [\[Paper\]](#)
- [9] Venelin Kovatchev, Trina Chatterjee, Venkata S Govindarajan, Jifan Chen, Eunsol Choi, Gabriella Chronis, Anubrata Das, Katrin Erk, Matthew Lease, Junyi Jessie Li, **Yating Wu**, Kyle Mahowald. “longhorns at DADC 2022: How many linguists does it take to fool a Question Answering model? A systematic approach to adversarial attacks.” DADC Workshop in The Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL Workshop**), 2022 [\[Paper\]](#)

Honors and Fellowships

Honors

- 2025 **EECS Rising Stars, MIT**
- 2024 **Outstanding Paper Award, EMNLP 2024**
- 2021 **1st Place in VMware Codehouse Palo Alto**
- 2019 **Outstanding Graduates of Dalian University of Technology**

Fellowships

- 2025 **Outstanding Student Lecture Series Fellowship, UT Austin**
- 2023-2024 **Professional Development Awards, UT Austin**

2022-2025 **Head TA fellowship, UT Austin**

Teaching Experience

Head Teaching Assistant

CS391L **Machine Learning (Graduate)**, Summer 2022 – Fall 2025

Instructors: Adam Klivans, Qiang Liu, UT Austin

Teaching Assistant

CS391L **Machine Learning (Graduate)**, Fall 2021 – Spring 2022

Instructors: Adam Klivans, Qiang Liu, UT Austin

Recitation Instructor

EE422C **Software Design and Implementation II (Undergraduate)**, Spring 2021

Independently designed and taught weekly recitation sections

EE422C **Software Design and Implementation II (Undergraduate)**, Summer 2020

Independently designed and taught weekly recitation sections

Mentoring Experience

Master's student **Ritika Mangla**, 2022-2024, co-authored paper [4] and [6], UT Austin

Master's student **Ashwin Ram**, 2024, UT Austin

Presentations

Talks

2025 Which Questions Should I Answer? Saliency Prediction of Inquisitive Questions, Outstanding Student Lecture Series, UT Austin

2024 Which Questions Should I Answer? Saliency Prediction of Inquisitive Questions, EMNLP 2024

2023 QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing, EMNLP 2023

Poster Presentations

2024 Elaborative Simplification as Implicit Questions Under Discussion, MLL Research Symposium, UT Austin

2023 Elaborative Simplification as Implicit Questions Under Discussion, EMNLP 2023

2022 Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion, ACL 2022

Industrial Presentation

2025 ContextWeaver: Selective and Interpretable Long Context Construction for LLM Agents, AWS AI Labs

2024 Self Adaptive Question Generation for Planning Agent, Amazon AGI

Service

Conference Organization

Workflow Chair, EMNLP 2026

Conference Reviewer

ACL Rolling Review (ARR) 2024-Now

ACL 2024, 2025, 2026

EMNLP 2024, 2025, 2026

COLM 2024, 2025, 2026

Workshops: ICML 2025 XAI4Science, AAAI 2025 MARW, NeurIPS 2024 Behavioral ML, ICML 2024 DMLR, ICML 2024 LLMs and Cognition

Department Service

Graduate student representative on faculty hiring panel

TA representative on online course infrastructure building

Technical Skills

AI/ML Pretraining/post-training, LLM/VLM evaluation, long-context agents, multi-modal AI for Science

Tools Python, PyTorch, Hugging Face, AWS, CUDA, Git